Check for updates

Pathogenesis of Post-Tuberculosis Lung Disease

Defining Knowledge Gaps and Research Priorities at the Second International Post-Tuberculosis Symposium

Sara C. Auld[1], Amy K. Barczak[2,3,4], William Bishai[5], Anna K. Coussens[6,7], Intan M. W. Dewi[8,9], Steven C. Mitini-Nkhoma[10], Caleb Muefong[11], Threnesan Naidoo[12,13,14], Anil Pooran[15,16], Cari Stek[17], Adrie J. C. Steyn[14,18,19], Liku Tezera[20], and Naomi F. Walker[21,22]

Abstract

Post-tuberculosis (post-TB) lung disease is increasingly recognized as a major contributor to the global burden of chronic lung disease, with recent estimates indicating that over half of TB survivors have impaired lung function after successful completion of TB treatment. However, the pathologic mechanisms that contribute to post-TB lung disease are not well understood, thus limiting the development of therapeutic interventions to improve long-term outcomes after TB. This report summarizes the work of the Pathogenesis and Risk Factors Committee for the Second International Post-Tuberculosis Symposium, which took place in Stellenbosch, South Africa, in April 2023. The committee first identified six areas with high translational potential: *1*) tissue matrix destruction, including the role of matrix metalloproteinase dysregulation and neutrophil activity; *2*) fibroblasts and profibrotic activity; *3*) granuloma fate and cell death pathways; *4*) mycobacterial factors, including pathogen burden; *5*) animal models; and *6*) the impact of key clinical risk factors, including HIV, diabetes, smoking, malnutrition, and alcohol. We share the key findings from a literature review of those areas, highlighting knowledge gaps and areas where further research is needed.

Keywords: tuberculosis; post-tuberculosis lung disease; fibrosis; neutrophils; matrix metalloproteinases

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

With over 10 million new cases and 1.6 million deaths each year, tuberculosis (TB) remains a major public health priority (1). However, morbidity and mortality caused by TB do not end after successful TB treatment. Rather, people treated for TB have a nearly threefold increased risk of death compared with people who have never had TB (2), and over half of pulmonary TB survivors have evidence of ongoing respiratory morbidity after completion of TB treatment (3). With an estimated 155 million TB survivors alive in 2020 (4), post-tuberculosis lung disease (PTLD) is a major contributor to the global burden of chronic lung disease, estimated to account for 47% of the 122 million disability-adjusted life-years attributed to incident TB in 2019 (5). Despite increasing recognition of the substantial epidemiologic burden of PTLD and growing consensus on clinical manifestations and phenotypes, from disease involving the airways and parenchyma to pleural and pulmonary vascular disease (Box 1), there is a paucity of knowledge regarding the pathologic mechanisms underlying PTLD. A greater understanding of these diverse mechanisms and how they correspond to distinct post-TB sequelae would enable the identification of biological subtypes, or endotypes, of PTLD that could be therapeutically targeted with directed interventions (6).

In April 2023, a multidisciplinary group of international stakeholders gathered for the Second International Post-Tuberculosis Symposium in Stellenbosch, South Africa (7). This report summarizes the work of the symposium's Pathogenesis and Risk Factors Committee, which aimed to identify key research priorities around biological pathways and mechanisms important to PTLD. The committee, composed of 13 investigators invited to participate on the basis of their clinical and scientific expertise, identified six areas with high translational potential for a focused review of the literature: *1*) tissue matrix destruction, including the role of matrix

(Received in original form February 17, 2024; accepted in final form August 13, 2024)

Supported by National Institute of Allergy and Infectious Diseases grants R01AI166988 (S.C.A.), R21AI146813 (A.K.B.), R37AI167750 (W.B.), and R33AI138280 (A.J.C.S.) and Australia National Health and Medical Research Council grant GNT2020750 (A.K.C.). The Second International Post-Tuberculosis Symposium was cofunded by the German Federal Ministry of Education and Research (01KA2223B) through the Research Networks for Health Innovations in Sub-Saharan Africa (RHISSA) TB-SEQUEL Network.

Correspondence and requests for reprints should be addressed to Sara C. Auld, M.D., M.S., Emory University, 615 Michael Street N.E., Suite 200, Atlanta, GA 30322. E-mail: sauld@emory.edu.

A data supplement for this article is available via the Supplements tab at the top of the online article.

Am J Respir Crit Care Med Vol 210, Iss 8, pp 979–993, Oct 15, 2024
Copyright © 2024 by the American Thoracic Society
Originally Published in Press as DOI: 10.1164/rccm.202402-0374SO on August 14, 2024
Internet address: www.atsjournals.org

[1]Departments of Medicine, Epidemiology, and Global Health, Emory University School of Medicine and Rollins School of Public Health, Atlanta, Georgia; [2]Division of Infectious Diseases, Massachusetts General Hospital, Boston, Massachusetts; [3]Ragon Institute of Massachusetts General Hospital, Massachusetts Institute of Technology, and Harvard, Cambridge, Massachusetts; [4]Harvard Medical School, Boston, Massachusetts; [5]Division of Infectious Diseases, Department of Medicine, Johns Hopkins School of Medicine, Baltimore, Maryland; [6]Infectious Diseases and Immune Defence Division, Walter and Eliza Hall Institute of Medical Research, Parkville, Victoria, Australia; [7]Department of Medical Biology, University of Melbourne, Parkville, Victoria, Australia; [8]Microbiology Division, Department of Biomedical Sciences, Faculty of Medicine, and [9]Research Center for Care and Control of Infectious Diseases, Universitas Padjadjaran, Bandung, Indonesia; [10]Malawi Liverpool Wellcome Clinical Research Programme, Blantyre, Malawi; [11]Department of Microbiology, University of Chicago, Chicago, Illinois; [12]Department of Forensic & Legal Medicine and [13]Department of Laboratory Medicine & Pathology, Faculty of Medicine & Health Sciences, Walter Sisulu University, Eastern Cape, South Africa; [14]Africa Health Research Institute, University of KwaZulu-Natal, Durban, South Africa; [15]Centre for Lung Infection and Immunity, Division of Pulmonology, Department of Medicine, and [16]University of Cape Town Lung Institute and Medical Research Council/University of Cape Town Centre for the Study of Antimicrobial Resistance, Cape Town, South Africa; [17]Wellcome Center for Infectious Diseases Research in Africa, Institute of Infectious Disease and Molecular Medicine, University of Cape Town, Cape Town, South Africa; [18]Department of Microbiology and [19]Centers for AIDS Research and Free Radical Biology, University of Alabama at Birmingham, Birmingham, Alabama; [20]National Institute for Health and Care Research Biomedical Research Centre, School of Clinical and Experimental Sciences, Faculty of Medicine, University of Southampton, Southampton, United Kingdom; [21]Department of Clinical Sciences and Centre for Tuberculosis Research, Liverpool School of Tropical Medicine, Liverpool, United Kingdom; and [22]Tropical and Infectious Diseases Unit, Liverpool University Hospitals NHS Foundation Trust, Liverpool, United Kingdom

ORCID IDs: 0000-0002-8972-3736 (S.C.A.); 0000-0003-3806-2381 (A.K.B.); 0000-0002-8734-4118 (W.B.); 0000-0002-7086-2621 (A.K.C.); 0000-0002-4910-9651 (I.M.W.D.); 0000-0002-3276-6890 (S.C.M.-N.); 0000-0001-8817-9188 (C.M.); 0000-0002-1864-4301 (T.N.); 0000-0001-5808-433X (A.P.); 0000-0001-8813-0819 (C.S.); 0000-0001-9177-8827 (A.J.C.S.); 0000-0002-7898-6709 (L.T.); 0000-0002-3345-7694 (N.F.W.).

Contents

Matrix Destruction, MMPs, and Neutrophils
Fibroblasts and Profibrotic Activity
Granuloma Fate and Cell Death Pathways
Mycobacterial Factors and Pathogen Burden
Experimental Models of PTLD
Interactions between Clinical Risk Factors and Pathogenesis of PTLD
HIV
Diabetes Mellitus
Undernutrition
Smoking and Alcohol Use Disorders
Research Priorities
Conclusions

metalloproteinase (MMP) dysregulation and neutrophil activity; *2*) fibroblasts and profibrotic activity; *3*) granuloma fate and cell death pathways; *4*) mycobacterial factors, including pathogen burden; *5*) animal models (Figure 1); and *6*) the impact of key clinical risk factors, including HIV, diabetes, smoking, malnutrition, and alcohol. (*See* Table E1 in the online supplement for a ranked list of research areas considered for review.) Here we report the key findings from this work and highlight knowledge gaps where further research focus is needed to inform and advance our understanding of the pathogenesis of PTLD. Although there is growing recognition of the many long-term sequelae of TB disease, from epigenetic modifications and alterations in the host microbiome to increased risks of cardiovascular disease and cancer (2, 8–11), this report focuses primarily on biological pathways and mechanisms relevant to the airways and lung parenchyma, which have been the most studied PTLD manifestations to date.

Matrix Destruction, MMPs, and Neutrophils

Destruction of the lung extracellular matrix during *Mycobacterium tuberculosis* (*Mtb*) infection often begins a considerable time before TB diagnosis and continues during TB treatment. The radiological hallmark of pulmonary TB is a cavity, radiologically defined as a lucency within a zone of pulmonary consolidation, a mass, or a nodule that is surrounded by a wall, classically affecting the lung apices (12). Cavitary TB is the major source of *Mtb* transmission and is associated with higher bacterial burden in respiratory secretions and longer time to microbiological cure (13); bronchiectasis often accompanies cavitation, and residual cavitation is a common feature of PTLD (14, 15). The absence of collagen and elastin, the major structural proteins of the lung, inside cavities points to the involvement of MMPs—a family of host enzymes that collectively degrade the extracellular matrix, including collagenases, gelatinases, and stromelysins. MMPs have roles in wound healing and are implicated in both tissue destruction and remodeling in TB (16). MMPs are physiologically tightly regulated but are upregulated in monocytes, macrophages, and neutrophils directly in response to *Mtb* infection and via cellular networks, including uninfected stromal cells, without compensatory upregulation of specific inhibitors, the tissue inhibitors of MMPs, described as a "matrix-degrading phenotype" (17–20). Elevated concentrations of MMPs, including MMP-1, -2, -3, and -8, have been detected in sputum from patients with TB compared with respiratory symptomatic and healthy control patients, and they have also been associated with cavitation and sputum smear positivity (21–24). MMP-1 polymorphisms have also been associated with more extensive pulmonary fibrosis and increased risk of tracheobronchial stenosis after TB treatment completion, further implicating this collagenase in both tissue damage and pathological remodeling leading to PTLD,

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

Box 1: Consensus definition and proposed clinical patterns of post-tuberculosis lung disease (PTLD) from the First International Post-Tuberculosis Symposium (155)

Minimum case definition of PTLD: Evidence of chronic respiratory abnormality, with or without symptoms, attributable at least in part to previous tuberculosis.

Proposed clinical patterns of PTLD:

1. Airways: TB-associated obstructive lung disease with airway obstruction on spirometry OR bronchiectasis on chest radiography
2. Parenchyma: Cavitation, parenchymal destruction, or fibrotic change on chest radiography OR aspergillus-related lung disease on imaging and/or blood testing
3. Pleural: Pleural thickening on chest imaging
4. Pulmonary vascular: Pulmonary hypertension with elevated pulmonary artery pressures on echocardiography and/or right heart catheterization
5. Other: Other pathology not meeting the above criteria

with pathological extracellular matrix remodeling considered an integral process in other fibrotic lung diseases, where MMP dysregulation has also been implicated, such as silicosis and idiopathic pulmonary fibrosis (IPF; *see below*) (25–29).

Although animal models are limited in their capacity to reproduce features of PTLD (*see below section*), they have provided specific mechanistic insights with respect to inflammation and matrix destruction during TB. For example, a long-standing tenet in TB pathology was that caseous necrosis led to tissue destruction and cavitation. However, a series of experiments leveraging transgenic MMP-1 expressing mice found that extracellular matrix destruction actually precedes caseous necrosis (30). This

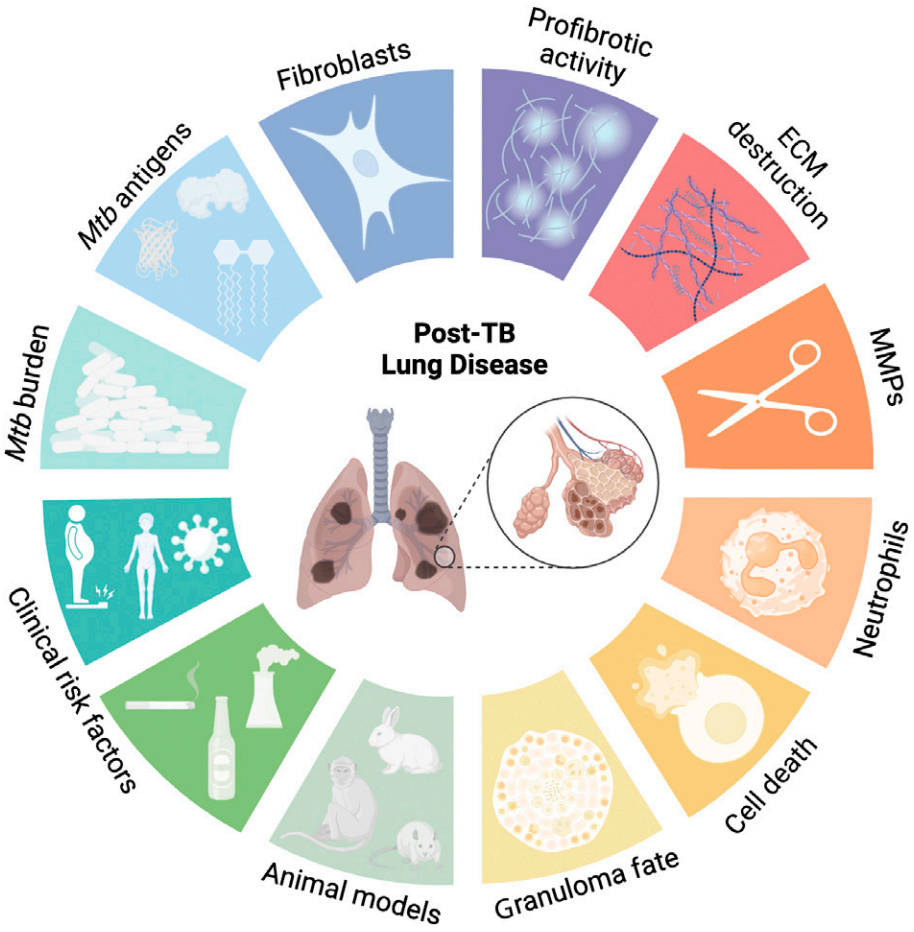


Figure 1. Consensus key areas of importance in post-tuberculosis lung disease pathogenesis. Created with BioRender.com. ECM = extracellular matrix; MMPs = matrix metalloproteinases; *Mtb* = *Mycobacterium tuberculosis*; TB = tuberculosis.

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

pathology, which is analogous to human TB, was not evident in wild-type or MMP-9–expressing transgenic mice (30). Animal models have also been used to study the effects of MMP inhibition in TB disease. The potent MMP-1 inhibitor cipemastat worsened TB pathology in the absence of concomitant TB treatment in a mouse model but did not impact cavitation in a rabbit model (31, 32). Combining MMP inhibition with TB treatment reduced tissue damage after *Mtb* infection in several other animal studies (reviewed in Reference 33). Administration of the broad MMP inhibitor marimastat promoted delivery and retention of rifampicin and isoniazid in the lungs of *Mtb*-infected mice because of an effect on the vasculature, with increased pericyte coverage and decreased leakage of surrounding blood vessels (34). In humans, a randomized controlled trial of doxycycline, a broad MMP inhibitor, administered for 2 weeks alongside TB treatment demonstrated that doxycycline reduced sputum MMP levels, suppressed collagen and elastin destruction, and decreased pulmonary cavity volume on chest radiograph (35). These findings provide the most robust evidence to date that MMP inhibition can mitigate matrix destruction in pulmonary TB and can potentially offer a strategy to reduce or prevent PTLD. However, these results should now be replicated in larger cohorts with extended follow-up, including measurement of lung function. Additional work is also needed to determine whether specific, more selective MMP inhibitors may be required, as has been seen in the cancer field (36, 37).

Neutrophils are innate phagocytes that may release MMPs and a range of inflammatory mediators in response to *Mtb*. Neutrophils have emerged as key players in TB immunity and immunopathology. Excessive neutrophil activity has been associated with the extent of inflammatory pathology and cavitation on chest radiographs, with a positive association reported between the peripheral blood neutrophil count and the extent of lung disease in patients with TB and increased concentrations of neutrophil-derived MMP-8, MMP-9, and S100A8 in patients with greater lung involvement on chest radiographs at baseline (38–40). In one study evaluating TB severity in humans, pulmonary destruction on chest radiographs, sputum *Mtb* load, disseminated TB, and death were strongly associated with a relative expansion of circulating neutrophils, particularly subtypes of segmented and banded neutrophils, and a decrease in lymphocyte/neutrophil ratios (41). Although chest radiograph severity at the time of diagnosis cannot be considered a reflection of PTLD, natural history studies indicate that the extent of radiographic disease at diagnosis is a strong predictor of the risk of PTLD, and baseline chest radiography has been used as a screening tool to identify individuals at high risk of PTLD in clinical trials (42, 43). One study has investigated neutrophil function by extent of radiographic recovery at TB treatment completion, finding baseline granulocyte proportions post-stimulation (with ESAT-6 [early secreted antigenic target 6]/CFP-10 [culture filtrate protein 10] fusion protein and *Mtb* lysate in a whole-blood assay) were increased in those with poor compared with good recovery (38).

In addition to the MMPs, evidence from patients and mouse studies suggests that neutrophil activity and products, including myeloperoxidase, neutrophil elastase (44), cytokines and chemokines (45), and even IFN-induced neutrophil extracellular trap (NET) formation (46), may contribute to lung damage in TB disease. *Mtb*-infected neutrophils were determined to produce NETs and colocalize with the collagenase MMP-8 *in vitro* (24). In patients with TB, NETs were noted in sputum, and MMP-8 was associated with cavitary disease on chest radiographs (24). Increased NET concentrations were also found in sputum from patients with TB as compared with healthy control subjects (24). Moreover, neutrophil-specific inflammatory mediators such as S100A8/9 and IL-8 have been associated with increased lung damage severity, with high *Mtb* loads and cavitation, in patients with TB pretreatment (38). In *Mtb*-infected C57BL/6 mice, IFN-γ neutralization resulted in exacerbated inflammation and increased accumulation of S100A8-producing Gr1 + neutrophils (47). S100A8/A9 production, which was induced by IL-17 overexpression, further promoted neutrophil accumulation by inducing production of proinflammatory chemokines and cytokines and influencing leukocyte trafficking. In contrast, IFN-γ neutralization in *Mtb*-infected S100A9 knockout mice resulted in loss of lung inflammation and reversed neutrophil accumulation, without impacting bacterial burdens. A separate study demonstrated that dysregulated granulocytic influx promotes disease by creating a permissive long-lived intracellular niche for mycobacterial growth and persistence (48). These studies suggest a disconnect between tissue damage and bacterial burden, with TB disease driven by inflammatory host factors rather than directly by the *Mtb* pathogen burden in the tissue. The complex relationship between Mtb pathogen burden and PTLD is discussed further below.

Fibroblasts and Profibrotic Activity

Pulmonary fibrosis is likely a substantial contributor to multiple clinical manifestations of PTLD, including reduced exercise capacity and chronic hypoxia (49–51). In a small cohort study of 30 people with pulmonary TB in India, elevated concentrations of the profibrotic cytokine transforming growth factor (TGF) and IL-6 were associated with worse lung function and greater respiratory symptoms, respectively, after TB treatment completion (52). Given the prevalence of TB globally and the prevalence of PTLD among TB survivors, TB is arguably the single leading cause of pulmonary fibrosis around the world. However, to date, there has been strikingly little investigation into TB-specific mechanisms of fibrogenesis. Rather, any understanding of TB-associated fibrosis is primarily inferred from studies of fibrosis in other pulmonary diseases. IPF is the most intensively studied pulmonary fibrotic process and serves as the basis for comparative understanding of fibrosis in other disease processes, including TB; however, the underlying pathologic drivers are likely quite different. The cells associated with fibrosis and molecules that contribute to fibrogenesis are of particular interest, because both offer potential therapeutic targets. In IPF, defined subsets of myofibroblasts have been associated with fibrogenesis, and endothelial cell injury and apoptosis have been implicated as early causative events (reviewed in [28]). A role for epithelial cell–derived secreted factors, including MMPs, in driving migration, activation, and proliferation of myofibroblasts and extracellular matrix remodeling has been proposed in IPF (28, 53). The technical approaches used to advance an understanding of molecular and cellular contributors to IPF pathogenesis, including

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

single-cell RNA sequencing and complementary immunofluorescence to define spatial relationships, hold significant promise for similarly advancing our understanding of TB-associated fibrosis (54, 55).

Focused studies in cellular and animal models of TB have offered some insights into TB-associated fibrogenesis. Although few cellular models are amenable to studying complex processes such as TB-associated fibrosis, exposure of human fibroblasts to *Mtb* results in the production of the secreted profibrotic connective tissue growth factor (CTGF) in a Toll-like receptor 2 (TLR2)-dependent fashion (56). To date, no small-animal models of TB have been applied to study the mechanisms of TB-associated fibrosis within tissue, but a few studies have used nonhuman primate (NHP) models. In one study, NHP granulomas were stratified on the basis of histopathologic staining for collagen to probe potential molecular associations with fibrosis. TGF and α-smooth muscle actin, two molecules implicated in fibrosis, were found to be associated with fibrotic TB granulomas (57). In a subsequent study, computational models validated with NHP specimens pointed to macrophage-to-fibroblast transformation as a key mechanism underlying granuloma-associated fibrosis in TB (58). Extension of that computational work suggested that the local cytokine milieu may contribute to the distinct patterns of fibrosis observed within granulomas with different spatial patterns of collagen deposition (59). The development of small-animal models that reflect human-like lung fibrotic pathology will enable mechanistic studies that identify candidate targets for antifibrosis therapies and preclinical studies of efficacy.

Ultimately, a full understanding of TB-associated fibrogenesis will depend upon tissue-centric studies in human lung samples. Studies taking advantage of more accessible compartments can offer insight into circulation of molecules known to be associated with fibrosis (60); however, understanding the spatial molecular and cellular relationships that drive fibrosis is necessary to develop future interventions. Although animal models can suggest what the cells, molecules, and relationships that drive TB-associated fibrosis might be, human tissue-based studies will be necessary to confirm or refute the relevance of predicted associations in human disease. Furthermore, such investigations must be performed using tissue samples from multiple time points during and after treatment of TB disease to enable an understanding of both early and late events in fibrogenesis and lung remodeling.

Granuloma Fate and Cell Death Pathways

TB granulomas embody a spectrum of disease states, featuring dynamic and spatially organized structures comprising macrophages, granulocytes, fibroblasts, and lymphocytes, among others. This intricate composition gives rise to a diverse spectrum of lesions, encompassing necrotic, nonnecrotic, caseous, fibrotic, cavitary, and calcified manifestations, whose fate is directly intertwined with the structural changes observed in PTLD. Although the necrotic granuloma is commonly perceived as the quintessential histopathological feature of *Mtb* infection, it is noteworthy that necrotic granulomas can also be induced by nontuberculous mycobacteria, sarcoidosis, fungal infections (e.g., histoplasmosis, cryptococcus, blastomycosis, coccidiomycosis), and certain vasculitides (e.g., granulomatosis with polyangiitis). The ultimate trajectory of the TB granuloma toward either effective healing or lung immunopathology associated with PTLD will depend on the type and extent of the host immune response, *Mtb* strain, and their interaction. A granuloma can effectively contain bacteria to prevent dissemination, as for those rich in inducible bronchus-associated lymphoid tissue (61). However, it can also serve as a reservoir for uncontrolled replication, as seen in neutrophil-rich granulomas (62).

Meanwhile, historical studies suggest that caseation leads to bacillary destruction, a proposition substantiated by the absence of acid-fast bacilli within the caseum, as demonstrated by Canetti in 1955 and as routinely observed in modern-day pathology laboratories (63). Although new, more sensitive techniques including RNAscope have been shown to detect *Mtb* RNA in lesions that are absent of stained acid-fast bacilli (64). As a hallmark of primary TB, caseating granulomas have been a considerable focus in animal models investigating the outcome of primary infection. However, after primary TB, the phenotype of disease that reflects most adult pulmonary TB is characterized by obstructive lobular pneumonia with abundant neutrophil infiltration (65). Secondary granulomas form as a consequence rather than as a cause of cavitation in post-primary TB.

Technological advances in the last several years have allowed unbiased approaches combining transcriptomic, proteomic, and imaging techniques to perform landscape analyses of diverse granulomas with a high degree of resolution. Techniques such as laser microdissection coupled with mass spectrometry (66, 67), immunofluorescence detection (68), multiplexed ion beam imaging by time of flight (69), single cell RNA sequencing (70), and micro–/nano–computed tomography (71, 72) have revealed tissue microenvironments within and between granulomas with distinct immunological signatures. A recent study in cynomolgus macaques investigating longitudinal granuloma development, which would be difficult to perform in humans, showed that early-appearing granulomas with a high bacterial burden were characterized by a type 2 antiinflammatory immune response, whereas late-appearing granulomas with a low bacterial burden had a T-helper type 1 cell (Th1)/Th17-mediated proinflammatory response (70). These studies also revealed extensive inter- and intralesion heterogeneity of granulomas within a single animal, which was highly influenced by the balance between proinflammatory and antiinflammatory/regulatory immune responses within the different tissue microenvironments. The ultimate course of the granuloma, whether it leads to successful healing or lung immunopathology and PTLD, hinges upon factors such as the nature and intensity of the host immune response to bacillary burden and antigens and the intricate interplay between the host and the pathogen. Further investigation with longitudinal sampling of granulomas during the course of disease is needed to determine the timing and molecular mechanisms that drive distinct granulomas and lung microenvironments toward these different PTLD manifestations.

The timing and triggering of different cell death pathways across cell types also influences initial granuloma fate and the phenotype and resolving or nonresolving nature of disease that develops (73). Apoptosis, primarily considered a protective noninflammatory response, will cause the infected cell to die and be cleared by

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

neighboring phagocytes and at the same time kill and degrade the intracellular bacilli trapped in the apoptotic cell, enhancing antigen presentation to T and B cells. *Mtb* has developed apoptosis-evasion mechanisms, tipping the cell to trigger various necrotic inflammatory cell death pathways that benefit the bacterium (74, 75). Inflammatory cell death releases alarmins, endogenous immunomodulatory molecules, to coordinate immune cell help, but rupture of the cell membrane also enables the release of both *Mtb* and cellular proteases that degrade tissue matrix.

Although *Mtb* has been shown to trigger a variety of inflammatory cell death programs, how these relate to the phenotypes of TB disease and their long-term impact on PTLD are relatively unknown. For example, ferroptosis, a cell death pathway associated with increased iron and lipid peroxidation, is induced only with high multiplicity of infection. Its relevance to human TB is not well established, but it may be that ferroptosis is a late response at states of high *Mtb* load and contributes to local tissue necrosis (76). Differential regulation of the inflammasome pathway that leads to pyroptosis has also been identified between clinical *Mtb* strains from patients with different degrees of involvement on chest radiographs, again raising the complex question of how mycobacterial factors may impact the risk of PTLD (77).

Mycobacterial Factors and Pathogen Burden

Although it is recognized that PTLD arises from intricate interactions between host, pathogen, and the environment, mycobacterial factors that may contribute to the development of PTLD have not been well studied. Rather than directly driving tissue injury, bacterial factors are likely to be contributing primarily through their interactions with the host. Nevertheless, multiple studies have reported associations between TB-associated lung injury, pathogen burden, the presence of drug resistance, and *Mtb* strain diversity.

One key question is precisely how host inflammatory responses to the bacterium contribute to acute lung injury and pathologic remodeling. Multiple *Mtb* products, including cell wall and envelope components trehalose-6,6′ dimycolate and lipomannan, are recognized by TLRs on host cells, with resulting release of proinflammatory cytokines (78). Furthermore, intracellular components such as heat shock proteins and bacterial nucleic acids interact with intracellular TLRs, further propagating inflammation (78). The *Mtb* ESX-1 secretion system is required for activation of multiple components of the host response to *Mtb* infection, including the type I IFN response and activation of AIM2 and NLRP3 inflammasomes with subsequent release of the proinflammatory cytokines IL-1 and IL-18 (79–81). Furthermore, secretion of ESX-1–dependent effectors ESAT-6 and CFP-10 inhibit macrophage phagolysosome fusion, induce IFN-γ production, activate T cells, and promote granuloma formation (78). ESX-1 proteins and the cell membrane component phthiocerol dimycocerosates also damage host cell membranes and promote cellular necrosis (82).

Recent work has identified a secreted *Mtb* effector, protein tyrosine phosphatase A, that translocates to the infected macrophage nucleus and suppresses expression of a ferroptosis and pyroptosis inhibitor, glutathione peroxidase-4, by regulating macrophage DNA methylation via protein arginine methyltransferase 6 (83). Another study found that protein tyrosine phosphatase B blocks pyroptosis by dephosphorylating host membrane lipids to prevent the pyroptosis effector cleaved gasdermin D from inserting into the plasma membrane (84). By limiting host clearance of *Mtb*, this inhibition of pyroptosis may contribute to prolonged inflammation, potentially increasing the risk of PTLD.

Exactly how each of these components of the inflammatory response then contributes to acute tissue injury and/or pathologic tissue remodeling is an important area for future investigation. The capacity of the bacterium to evade sterilization by the host immune system, resulting in progressive immune cell recruitment and inflammatory signaling cascades at sites of infection, is central to the TB disease process. *Mtb* factors that drive host cell death and enable bacterial survival in host cells, including phthiocerol dimycocerosates and ESX-1, might thus be similarly considered to contribute to PTLD. However, despite the critical role for these proteins and lipids in *Mtb* infection and pathogenesis, their role in PTLD has not been studied. Animal models that enable quantification of individual components of PTLD are necessary to systematically and rigorously determine the contribution of individual bacterial factors to post-treatment features such as fibrosis, bronchiectasis, and residual cavitation.

There are conflicting data on the association between *Mtb* strain lineage and patterns of lung injury in patients with TB. For example, although the *Mtb* lineage 2 was associated with cavitary TB in one study from Russia, another study from Singapore found that lineage 2 actually had a lower propensity to cause cavitation than other strains (85, 86). These conflicting reports highlight the complexity of the interactions between *Mtb* and the host immune response, including whether these interactions may be impacted by different sublineages, different host populations and genomics, and different TB treatment regimens. Using a C3HeB/FeJ mouse model, Verma and colleagues demonstrated that "high-transmission *Mtb* strains," which were collected from individuals where a high proportion (⩾70%) of TB household contacts had a positive tuberculin skin test result, had a lower bacterial burden but developed caseating granulomas with high potential to cavitate (87). In contrast, mice infected with "low-transmission *Mtb* strains," collected from individuals where a low proportion (⩽40%) of household contacts had a positive tuberculin skin test result, exhibited significantly higher total lung bacterial burden with higher intracellular *Mtb* replication, developed diffuse inflammatory lung pathology and high neutrophil influx, and induced significantly higher lipid droplet formation in macrophages (87). During the chronic stages of infection, mice infected with low-transmission strains showed widespread tissue destruction, whereas those infected with high-transmission strains had expanding, well-formed granulomas and higher extracellular numbers of *Mtb* bacilli despite a lower total *Mtb* burden (87). These data suggest that strains with different transmission potential may also induce distinct pulmonary pathology. Further studies are needed to determine whether strain-level factors, and how those factors elicit particular aspects of the host immune response, impact the development of specific post-TB pathologic sequelae.

Insight into the impacts of mycobacterial strain on the risk of PTLD may also be garnered from a comparison of *Mtb* and *Mycobacterium africanum* (*M. africanum*), an important cause of TB

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

disease in parts of West Africa. Although there are no published studies specifically focused on post-TB sequelae after *M. africanum* disease, there are conflicting data regarding whether the extent of radiographic disease and cavitation at the time of diagnosis is higher or lower with *M. africanum* than with *Mtb* (88, 89). In addition, both the response to standard TB treatment regimens and the host immune response may differ for *M. africanum*, both of which could plausibly influence the risk of PTLD (90–92). Studies directly comparing PTLD rates and features between *Mtb* and *M. africanum* could yield important insights into how strain and lineage contribute to PTLD.

The causative relationship between *Mtb* burden and PTLD remains uncertain. Although there are limited data to shed light on this question, several studies have found associations between *Mtb* burden and histologic and radiologic changes in patients with TB and animal models, including positive associations between bacterial burden and fibrosis within granulomas in both macaques and humans (70, 93). *Mtb* burden detected in sputum in human disease may more likely reflect the extent of lung damage facilitating the translocation of *Mtb* bacilli into the airways, and *Mtb* growth in the wall of the cavity (94), than the total tissue *Mtb* burden. Thus, it remains unclear if high *Mtb* burden is a cause or a consequence of fibrotic or cavitary lung lesions, and the status of the host immune response is likely central to this relationship (*see* Figure 2).

There are also limited data on the association between drug resistance and the risk of PTLD. A recent cross-sectional study from Peru reported that survivors of multidrug-resistant TB are more likely to have a restrictive pattern on spirometric testing than survivors of drug-susceptible TB (95). However, such findings likely reflect diagnostic delays for drug-resistant disease and/or the prolonged *Mtb* survival in tissues in the setting of less effective drug regimens, both of which would prolong the local inflammatory response and ongoing tissue damage rather than the drug resistance itself. The development of preclinical models for PTLD, discussed below, will enable investigation of whether differences in mycobacterial strain and early bactericidal activity of antimycobacterial regimens impact the risk of PTLD. Adding measures of pulmonary impairment (e.g., chest radiography, spirometry) to ongoing clinical trials of novel TB treatment regimens will also provide critical human data on the role of ongoing bacterial replication in driving the risk of PTLD.

Experimental Models of PTLD

Animal models can play a crucial role in enhancing our understanding of PTLD. Mechanistic and preclinical studies will ultimately require validated preclinical models of PTLD. Although PTLD has multiple component pathologies, to date, animal models have mostly been used to study tissue destruction, as described for MMPs and neutrophils above, and granuloma formation and fate. Although mice are commonly used in studies of TB pathogenesis and offer logistical advantages for mechanistic studies, wild-type mice do not naturally develop necrotic tuberculous lesions, limiting their use in studying PTLD. Nonetheless, the C3HeB/FeJ and B6.Sst1S models develop necrotic granulomas resembling human TB lesions (96). Although C3HeB/FeJ mice harbor multiple *Mtb* susceptibility loci, one of which is the Sst1^{S} locus, the B6.Sst1^{S} mouse model possesses only the Sst1^{s} locus, which contains

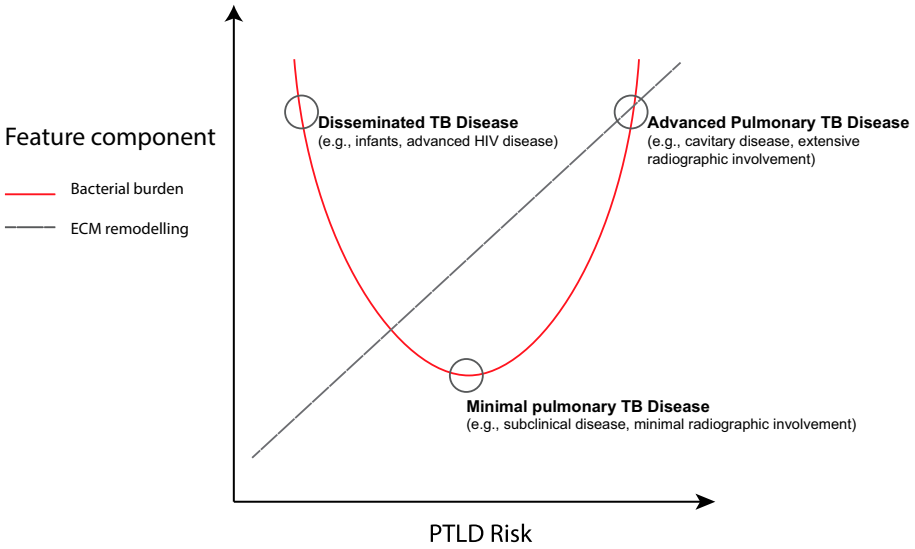


Figure 2. Proposed relationship between *Mycobacterium tuberculosis* (*Mtb*) bacterial burden, extracellular matrix (ECM) remodeling, and risk of post-tuberculosis lung disease (PTLD). Disseminated tuberculosis (TB) disease often occurs in the setting of immunodeficiency (e.g., advanced HIV) or a less developed immune system (e.g., infants) and is characterized by less clinically evident ECM damage and remodeling. In disseminated disease, sputum *Mtb* burden may be low because of a lack of cavitation, but there is a high *Mtb* burden in tissues on histopathological analysis and/or postmortem findings, and mortality is typically high. In patients with minimal pulmonary disease, there is typically a low *Mtb* burden with moderate pulmonary immune responses and accompanying ECM damage remodeling, in addition to a relatively lower risk of PTLD. In advanced pulmonary disease, including classical post-primary TB with upper lobe cavitation, there is a high *Mtb* burden, often with accompanying sputum smear acid-fast bacilli positivity, and more extensive ECM damage and remodeling, with a greatly increased risk of PTLD.

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

mutations in the SP110 and SP140 genes, backcrossed onto the C57BL/6 genetic background (97, 98). Macrophages from B6.Sst1^S mice release high levels of type I IFN in response to TNF, leading to a proteotoxic stress response and upregulation of the integrated stress response (ISR) (99). Inhibiting the ISR pathway with the small-molecule inhibitor ISRIB reduces bacterial load and necrotic lung pathology in B6.Sst1S mice (100). Similarly, the IL-13–overexpressing transgenic mouse develops human-like necrotic granulomas after *Mtb* infection (101, 102). Granulomas in both of these models behave similarly to human lesions, including their ability to restrict access to certain anti-TB drugs. These mouse models provide valuable insights into causal mechanisms that may underlie lung tissue damage leading to PTLD but require further validation to determine which pathologic features of PTLD are reliably reproduced in each model.

Other animal models have also been used for modeling TB disease and have characteristics that may be important for gaining insight into PTLD. The rabbit model, which can develop cavitary lung lesions, has yielded several essential insights regarding how cavities and their scar influence lung function (13, 103). The NHP model, using either cynomolgus or rhesus macaques, is the most human-like animal model for TB infection and disease. Upon intrabronchial instillation of *Mtb*, approximately 50% of cynomolgus macaques develop progressive lung disease with various features, including lymphadenopathy, granulomas, cavitation, and tuberculous pneumonia (104, 105). However, even with the ability to control for *Mtb* strain during inoculation, there remains substantial heterogeneity in the spectrum of pulmonary involvement, and eliciting consistent disease outcomes remains challenging.

To date, animal model studies of TB have not traditionally incorporated measures and outcomes specifically focused on PTLD. However, a recent report describes the development of a mouse model to quantify fibrosis during and after TB treatment, opening the door to much-needed mechanistic studies (106). Ideally, such studies would incorporate multiple, complementary measures, including pathogen factors, histopathology, radiographic assessments, and dynamic measures of lung function, which are increasingly available, even for small-animal models (107). Furthermore, these outcomes could be assessed at multiple time points during and after TB treatment to provide longitudinal insights into the trajectory of lung damage, remodeling, and recovery, which could be paired with multiomic measures of host responses (*See* Research Priorities).

Given the limited availability of applicable *in vivo* models, there is also a need to develop *in vitro* models for studying PTLD (108). An ideal *in vitro* model would use primary human cells and replicate the complex three-dimensional, multicellular environment of the lung while also reflecting the physiologic and cellular characteristics of TB-affected human lungs (109). Three-dimensional lung matrix models and airway organoids have been used to study immune responses in TB but do not yet model long-term outcomes (110, 111). These preclinical models, however, have the potential to be used for studying specific pathogen factors and toxicity of novel therapeutics, thus speeding up the discovery and evaluation of new host-directed therapies. Incorporating fibrotic elements or matrix-producing cell types would make it possible to experimentally mimic the pathological changes observed in PTLD and investigate host–pathogen interactions. The integration of high-throughput methodologies would also facilitate the rapid screening of potential therapeutic agents to prevent or mitigate the development of PTLD.

Interactions between Clinical Risk Factors and Pathogenesis of PTLD

There are substantial gaps in our understanding of how the five key risk factors for TB disease—undernourishment, HIV infection, alcohol use disorders, smoking, and diabetes mellitus—impact the pathogenesis and molecular pathways associated with the development of PTLD (1). Rather, many studies have examined the association between clinical risk factors and the extent of radiographic involvement at TB diagnosis, which has, in turn, been associated with the risk of PTLD (43, 112). There is a similar paucity of information about the impact of other key clinical risk factors for TB, including indoor and outdoor air pollution and occupational exposures such as silica dust, on the long-term risk for PTLD (113–115). It is also important to recognize that these clinical risk factors are often associated with social determinants of health, which could result in delayed care-seeking behavior and delays in TB diagnosis, which would likely increase the extent of lung damage before treatment initiation.

HIV

Although HIV is the greatest individual risk factor associated with TB, it remains uncertain how HIV may impact the risk of PTLD. A cross-sectional study conducted at community health clinics in Nigeria found that people with HIV had an increased risk of chronic lung disease (defined by either self-report or abnormal spirometry) in models adjusted for a history of TB (adjusted odds ratio [aOR], 2.35; 95% confidence interval [CI], 1.33–4.17), reflecting a greater risk for PTLD in people with HIV (116). However, in a similar analysis of people with and without HIV in western Kenya, HIV infection was not associated with impaired spirometry after adjustment for a history of TB (aOR, 1.1; 95% CI, 0.60–1.9), nor was HIV coinfection associated with symptomatic PTLD (aOR, 1.1; 95% CI, 0.4–2.9) in a cohort study from Tanzania (117, 118). Furthermore, in a longitudinal study of TB survivors, HIV coinfection was associated with higher lung function 3 years after treatment completion (119). However, this association was no longer significant after adjustment for radiologic extent of disease at TB treatment completion, suggesting that people with and without HIV with a similar extent of lung involvement may have a similar risk of PTLD. No studies have explicitly examined the association between CD4 count and PTLD, but CD4 count has been linked to the extent of radiographic disease at diagnosis (40, 120), indicating that individuals with a higher CD4 count may be at greater risk for PTLD. Of note, for the studies in Nigeria, Kenya, and Malawi, more than 80% of the people with HIV were receiving antiretroviral therapy and/or had an undetectable viral load. Although few studies have explicitly examined the pathogenesis of PTLD according to HIV status, higher plasma levels of MMP-8, but not MMP-9, were seen in patients with TB/HIV coinfection than in those with TB alone in a study in Indonesia (121). Furthermore, in two studies of people with and without HIV with pulmonary TB in South Africa, baseline peripheral blood neutrophil count and circulating IL-6 and IL-1RA expression were associated with the

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

extent of chest radiograph involvement and cavitation, respectively, but these associations were independent of HIV status (40, 122).

Diabetes Mellitus

The epidemiologic association between TB and diabetes mellitus is well established, including the greater risk of poor treatment outcomes and greater mortality both during and after TB treatment completion among those with TB and diabetes (123–125). In a study from Peru, people with TB who had dysglycemia had significantly more cavities, alveolar infiltrates, and fibrous tracts at the time of TB diagnosis, and the odds of abnormal radiographic findings increased per unit increase in HbA1c (126). A recent review of the converging epidemics of TB and diabetes by Ronacher and colleagues focused primarily on susceptibility to TB and impaired antimycobacterial immunity for people with diabetes, but they also discussed the altered inflammatory milieu in diabetes (127). Such alterations range from lower frequencies of *Mtb*-specific T cells and higher Th1 and Th17 cytokines to the accumulation of advanced glycation end products and reactive oxygen species, all of which have implications for the risk of lung damage, delayed healing, and PTLD. Likewise, a series of studies by Kumar and colleagues conducted in India identified a number of features of TB disease among those with diabetes that might increase the risk for PTLD, including *1*) delayed resolution of systemic inflammation at the end of TB treatment (128); *2*) upregulation of inflammatory lipid eicosanoids during and after TB treatment (129); *3*) higher levels of multiple MMPs at the time of TB diagnosis for people with diabetes, which were also significantly correlated with HbA1c (130); and *4*) higher levels of the inflammatory mediators advanced glycation end products, sRAGE (soluble receptor for advanced glycation end products), and the neutrophil-derived protein S100A12 in people with diabetes and TB that persisted throughout TB treatment (131). Another study from India found both greater correlations between proinflammatory lipid profiles and greater network connectivity between proinflammatory and proresolving lipid mediators in those with TB and diabetes as compared with those with TB alone (132). In the context of PTLD, such greater connectivity may have implications for the trajectory of lung damage and remodeling over the course of TB treatment.

Undernutrition

Although no studies have investigated the relationship between undernutrition and PTLD, several studies have found an association between undernutrition, typically defined as a low body mass index, and the extent of chest radiograph involvement or the presence of cavitation at the time of TB diagnosis (133–136). Low body mass index has also been associated with poor treatment outcomes of treatment failure, relapse, and death (137). A recent randomized trial in India found that nutritional supplementation led to reduced mortality for patients with TB and a reduced incidence of TB among their household contacts, but outcomes relevant to PTLD were not reported, so it is unknown whether nutritional supplementation would be associated with less chest radiograph involvement at the time of TB diagnosis or better lung function after TB treatment completion (138, 139). Finally, in a cross-sectional study of people with PTLD in Nigeria, weight loss was associated with poor health-related quality of life (140).

Smoking and Alcohol Use Disorders

A recent systematic review and meta-analysis found that tobacco use increased the risk of poor TB treatment outcomes, including recurrence, relapse, and death during treatment (141). With regard to the risk of PTLD and smoking, prior TB was associated with airflow limitation even after adjustment for smoking in a nationally representative survey from Korea (142). There was also evidence of statistical interaction between smoking and previous TB on airflow limitation, with a greater risk of airflow limitation according to the number of cigarette pack-years smoked (142). In addition, in a prospective cohort study in Mozambique, a greater than 10–pack-year smoking history was associated with a greater risk of PTLD (relative risk, 7.5; $P = 0.06$) (143). Finally, in a study from Cameroon, inflammatory cytokine levels in sputum were lower among those with PTLD than in those with smoking-related chronic obstructive pulmonary disease (144), suggesting that people with airflow obstruction after successful TB treatment may have less persistent inflammation in the lungs than those with ongoing smoke exposure.

In contrast to the growing literature supporting associations between smoking at PTLD, there are no studies examining the impact of alcohol use disorders on PTLD. However, as with other clinical risk factors, multiple studies have reported an association between alcohol use disorders and both cavitation and radiographic involvement at the time of TB diagnosis and poor TB treatment outcomes (145–147), all of which would point to a higher risk for PTLD.

Research Priorities

Despite the progress made in understanding matrix destruction in TB, many knowledge gaps with respect to the pathogenesis of PTLD still exist (Table 1). There is a need for small-animal models that better represent and recapitulate key features of human pulmonary TB and PTLD, including matrix destruction, cavitation, and fibrosis, and the extent of residual disease after treatment (Table 2). With such models, the contributions of individual molecules and cellular subsets in either matrix destruction and/or pathologic repair can be established. Determining which cell death pathways in which cells lead to which pathology, how bacterial factors and different *Mtb* strains regulate these pathways, and whether different timing in skewing toward certain cell death pathways leads to different outcomes will also be critical to understanding the contributors to necrotic tissue destruction in TB. Likewise, models that allow the evaluation of key comorbidities will help elucidate how various molecular mechanisms may be impacted by clinical risk factors known to exacerbate disease severity (e.g., HIV coinfection, diabetes mellitus, undernutrition, alcohol use, and cigarette smoking). In the case of the NHP model, given that cynomolgus macaques are known to develop many features of human pulmonary TB, repeated lung function measures should be included in NHP studies of TB pathogenesis and treatment. The development and validation of these small- and large-animal models will ultimately enable the development and preclinical testing of novel therapeutics designed to prevent or reduce PTLD.

Ultimately, any findings from animal models will need to be validated in humans. In-depth analysis of human lung tissue samples from people with TB disease, whether from surgical resections or autopsy samples, will help to confirm the relevance of any preclinical findings in human disease. The addition of endpoints relevant to PTLD in both observational cohorts and clinical

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

Table 1. Major Findings and Research Priorities for Pathogenesis of Post-TB Lung Disease

Topic Area	Major Findings to Date		
	Findings from Animal Models of TB	**Findings from Human Studies of TB**	**Findings Specific to Post-TB Lung Disease**
Matrix destruction, MMPs, and neutrophils	• Cipemastat (MMP-1 inhibitor) worsened TB pathology in the absence of anti-TB treatment in a mouse model • Combining MMP inhibition with TB treatment reduced tissue damage • Marimastat (broad MMP inhibitor) improved delivery and retention of anti-TB drugs in the lungs of *Mtb*-infected mice via effects on vasculature • IL-17 implicated in neutrophil accumulation through S100A8/9 and proinflammatory cytokine networks, regulated by IFN-γ	• Elevated MMP-1, -2, -3, and -8 in sputum, associated with cavitation • Neutrophil abundance, phenotype, and activity associated with extracellular matrix destruction and immunopathology • MMPs, advanced glycation end products, sRAGE, and neutrophil-derived S100A12 levels higher at the time of TB diagnosis in people with diabetes mellitus; MMP levels correlated with HbA1c • Adjunctive doxycycline (MMP inhibitor) with TB treatment reduced sputum MMPs, suppressed collagenase activity, and decreased cavity volume	• MMP-1 polymorphisms associated with more extensive pulmonary fibrosis and increased risk of tracheobronchial stenosis after TB treatment completion
Fibroblasts and profibrotic activity	• Exposure of human fibroblasts to *Mtb* results in profibrotic connective tissue growth factor secretion in a TLR2-dependent fashion • In nonhuman primates, TGF-β and α-SMA were associated with fibrotic granuloma formation following *Mtb* infection; macrophage-to-fibroblast transformation and local cytokine milieu implicated by computational modeling	—	• Elevated levels of TGF-β and IL-6 associated with lower lung function and more symptoms after TB treatment completion
Granuloma fate and cell death pathways	• Interplay of pathogen and host factors within granuloma, including cell death pathways, determines *Mtb* elimination, control, or disease • In NHP models, timing of granuloma formation after *Mtb* infection corresponds to *Mtb* burden and cytokine profile	• Radiographic severity of TB linked to differential regulation of inflammasome pathway and pyroptosis	—
Mycobacterial factors and pathogen burden	• Multiple *Mtb*-derived molecules can trigger host inflammatory responses, including trehalose-6,6′ dimycolate, lipomannan, and bacterial nucleic acids • Early secreted antigenic Target 6, culture filtrate protein 10) and phthiocerol dimycocerosates damage host cell walls and promote cellular necrosis	• High bacterial burden inside a TB cavity is reflected in higher *Mtb* loads in sputum in cavitary disease • Conflicting data on association between *Mtb* strain and risk of cavitary disease	• MDR-TB survivors in Peru more likely to have a restrictive pattern on spirometry than survivors of drug-susceptible TB

Definition of abbreviations: α-SMA = α-smooth muscle actin; MDR-TB = multidrug-resistant tuberculosis; MMP = matrix metalloproteinase; *Mtb* = *Mycobacterium tuberculosis*; TB = tuberculosis; TGF-β = transforming growth factor-β.

trials will also enhance our understanding of the clinical risk factors and biomarkers associated with PTLD and allow the definition of both clinical phenotypes and biological endotypes of PTLD. Such studies would require time points extending beyond TB treatment completion to shed light on how lung remodeling may continue to evolve even after microbiologic cure. It will be equally important to include pediatric cohorts in these studies, because recent data indicate that children who complete TB

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

Table 2. Research Priorities and Proposed Strategies

Overarching priorities	• Consensus and standardization of endpoints to quantify and classify the extent, course, and nature of matrix damage in both animal models and humans • Longitudinal studies with extended follow-up periods beyond completion of TB treatment, with sampling at multiple time points to investigate early and late events associated with PTLD in both preclinical and clinical studies
Preclinical model systems	• Development of new cellular and animal models to reflect the full spectrum of human disease, including cavitation and fibrosis, and with inclusion of post-treatment outcomes • Adoption of cellular and animal models developed for studying other pulmonary diseases (e.g., IPF, pulmonary hypertension) and assessing their utility for PTLD • Inclusion of physiologic measures of lung function in animal model studies • Evaluation of mechanisms of bacterial invasion and dissemination that lead to prolonged host inflammatory responses • Investigation of mechanisms implicated in granuloma trajectory toward *Mtb* clearance or immunopathology • Identification of *Mtb* mutants that increase lung pathology • Development of advanced *in vitro* models using primary human cells to replicate the complex three-dimensional environment of the lung • Use of high-throughput multiparameter readout platforms for breadth and efficiency • Inhibition studies to identify key pathways amenable to intervention and host-directed therapies • Simulation of risk factors in model systems (e.g., dysglycemia and HIV coinfection)
Human cohort studies and interventional trials	• Incorporation of translational and mechanistic substudies within ongoing clinical cohorts • Use of advanced imaging modalities (e.g., PET, hyperpolarized gas MRI) to gain insight into cellular metabolism and processes in clinical cohorts • Collection of lung tissue samples from sites of TB disease and from people with PTLD • Collection of *Mtb* isolates to enable analysis of the relationship between pathogen burden, strain, and resistance profiles with post-TB sequelae • Cellular and tissue-based studies with samples from disease sites, including advanced high-throughput sequencing and multiomic analyses • Larger randomized trials of MMP inhibition (and other host-directed strategies) to prevent or reduce PTLD

Definition of abbreviations: IPF = idiopathic pulmonary fibrosis; MMP = matrix metalloproteinase; MRI = magnetic resonance imaging; *Mtb* = *Mycobacterium tuberculosis*; PET = positron emission tomography; PTLD = post-tuberculosis lung disease; TB = tuberculosis.

treatment also experience post-TB morbidity and reduced lung function (148). The incorporation of advanced imaging modalities has the potential to provide further insight into both cellular metabolism and activity, with positron emission tomographic imaging using fluorodeoxyglucose or novel tracers targeting type 1 collagen (149–151), and gas exchange in the lung, with hyperpolarized xenon magnetic resonance imaging (152). Funders of TB treatment trials need to acknowledge that extended follow-up is essential and allocate funds accordingly. Efforts to establish consensus around standardized measures and definitions for PTLD to facilitate comparisons between studies are underway (153). Validation of these definitions is an important next step in this process.

Conclusions

Given increasing recognition of PTLD as a major contributor to global pulmonary morbidity, a greater understanding of the underlying pathogenesis is necessary to identify therapies to improve long-term outcomes for TB survivors. Yet, large gaps exist in our understanding of the biological mechanisms underlying PTLD and how key clinical risk factors influence those mechanisms and the accompanying long-term risk of PTLD. Addressing these gaps will require concerted efforts from researchers, clinicians, policymakers, and funders to identify synergies that can be gained by extending existing studies of TB pathogenesis with the addition of clinical endpoints relevant to PTLD. Such studies, in both animal models and humans, will be critical to understand how best to prevent excessive matrix damage, aberrant remodeling, and fibrosis in people with pulmonary TB. Over the last decade, we have witnessed significant progress in the prevention, diagnosis, and treatment of TB (154); it is now time to meaningfully advance our understanding of PTLD. ■

Author disclosures are available with the text of this article at www.atsjournals.org.

References

1. World Health Organization. Global tuberculosis report 2022. Geneva, Switzerland: World Health Organization; 2022.
2. Romanowski K, Baumann B, Basham CA, Ahmad Khan F, Fox GJ, Johnston JC. Long-term all-cause mortality in people treated for tuberculosis: a systematic review and meta-analysis. *Lancet Infect Dis* 2019;19:1129–1137.
3. Taylor J, Bastos ML, Lachapelle-Chisholm S, Mayo NE, Johnston J, Menzies D. Residual respiratory disability after successful treatment of pulmonary tuberculosis: a systematic review and meta-analysis. *EClinicalMedicine* 2023;59:101979.

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

4. Dodd PJ, Yuen CM, Jayasooriya SM, van der Zalm MM, Seddon JA. Quantifying the global number of tuberculosis survivors: a modelling study. *Lancet Infect Dis* 2021;21:984–992.
5. Menzies NA, Quaife M, Allwood BW, Byrne AL, Coussens AK, Harries AD, *et al.* Lifetime burden of disease due to incident tuberculosis: a global reappraisal including post-tuberculosis sequelae. *Lancet Glob Health* 2021;9:e1679–e1687.
6. DiNardo AR, Nishiguchi T, Grimm SL, Schlesinger LS, Graviss EA, Cirillo JD, *et al.* Tuberculosis endotypes to guide stratified host-directed therapy. *Med* 2021;2:217–232.
7. Allwood BW, Nightingale R, Agbota G, Auld S, Bisson GP, Byrne A, *et al.* Perspectives from the 2nd International Post-Tuberculosis Symposium: mobilising advocacy and research for improved outcomes. *IJTLD Open* 2024;1:111–123.
8. DiNardo AR, Rajapakshe K, Nishiguchi T, Grimm SL, Mtetwa G, Dlamini Q, *et al.* DNA hypermethylation during tuberculosis dampens host immune responsiveness. *J Clin Invest* 2020;130:3113–3123.
9. Scriba TJ, Carpenter C, Pro SC, Sidney J, Musvosvi M, Rozot V, *et al.* Differential recognition of *Mycobacterium tuberculosis*-specific epitopes as a function of tuberculosis disease history. *Am J Respir Crit Care Med* 2017;196:772–781.
10. Basham CA, Smith SJ, Romanowski K, Johnston JC. Cardiovascular morbidity and mortality among persons diagnosed with tuberculosis: a systematic review and meta-analysis. *PLoS One* 2020;15:e0235821.
11. Luczynski P, Poulin P, Romanowski K, Johnston JC. Tuberculosis and risk of cancer: a systematic review and meta-analysis. *PLoS One* 2022; 17:e0278661.
12. Hansell DM, Bankier AA, MacMahon H, McLoud TC, Müller NL, Remy J. Fleischner Society: glossary of terms for thoracic imaging. *Radiology* 2008;246:697–722.
13. Urbanowski ME, Ordonez AA, Ruiz-Bedoya CA, Jain SK, Bishai WR. Cavitary tuberculosis: the gateway of disease transmission. *Lancet Infect Dis* 2020;20:e117–e128.
14. Meghji J, Simpson H, Squire SB, Mortimer K. A systematic review of the prevalence and pattern of imaging defined post-TB lung disease. *PLoS One* 2016;11:e0161176.
15. Long R, Maycher B, Dhar A, Manfreda J, Hershfield E, Anthonisen N. Pulmonary tuberculosis treated with directly observed therapy: serial changes in lung structure and function. *Chest* 1998;113:933–943.
16. Elkington PT, Ugarte-Gil CA, Friedland JS. Matrix metalloproteinases in tuberculosis. *Eur Respir J* 2011;38:456–464.
17. Singh S, Saraiva L, Elkington PT, Friedland JS. Regulation of matrix metalloproteinase-1, -3, and -9 in *Mycobacterium tuberculosis*-dependent respiratory networks by the rapamycin-sensitive PI3K/p70(S6K) cascade. *FASEB J* 2014;28:85–93.
18. Elkington PT, Nuttall RK, Boyle JJ, O'Kane CM, Horncastle DE, Edwards DR, *et al. Mycobacterium tuberculosis*, but not vaccine BCG, specifically upregulates matrix metalloproteinase-1. *Am J Respir Crit Care Med* 2005;172:1596–1604.
19. Kübler A, Luna B, Larsson C, Ammerman NC, Andrade BB, Orandle M, *et al. Mycobacterium tuberculosis* dysregulates MMP/TIMP balance to drive rapid cavitation and unrestrained bacterial proliferation. *J Pathol* 2015;235:431–444.
20. Price NM, Farrar J, Tran TT, Nguyen TH, Tran TH, Friedland JS. Identification of a matrix-degrading phenotype in human tuberculosis in vitro and in vivo. *J Immunol* 2001;166:4223–4230.
21. Walker NF, Wilkinson KA, Meintjes G, Tezera LB, Goliath R, Peyper JM, *et al.* Matrix degradation in human immunodeficiency virus type 1-associated tuberculosis and tuberculosis immune reconstitution inflammatory syndrome: a prospective observational study. *Clin Infect Dis* 2017;65:121–132.
22. Sathyamoorthy T, Sandhu G, Tezera LB, Thomas R, Singhania A, Woelk CH, *et al.* Gender-dependent differences in plasma matrix metalloproteinase-8 elevated in pulmonary tuberculosis. *PLoS One* 2015;10:e0117605.
23. Walker NF, Clark SO, Oni T, Andreu N, Tezera L, Singh S, *et al.* Doxycycline and HIV infection suppress tuberculosis-induced matrix metalloproteinases. *Am J Respir Crit Care Med* 2012;185:989–997.
24. Ong CW, Elkington PT, Brilha S, Ugarte-Gil C, Tome-Esteban MT, Tezera LB, *et al.* Neutrophil-derived MMP-8 drives AMPK-dependent matrix destruction in human pulmonary tuberculosis. *PLoS Pathog* 2015;11:e1004917.
25. Wang CH, Lin HC, Lin SM, Huang CD, Liu CY, Huang KH, *et al.* MMP-1(-1607G) polymorphism as a risk factor for fibrosis after pulmonary tuberculosis in Taiwan. *Int J Tuberc Lung Dis* 2010;14:627–634.
26. Kuo HP, Wang YM, Wang CH, He CC, Lin SM, Lin HC, *et al.* Matrix metalloproteinase-1 polymorphism in Taiwanese patients with endobronchial tuberculosis. *Tuberculosis (Edinb)* 2008;88:262–267.
27. Ganachari M, Ruiz-Morales JA, Gomez de la Torre Pretell JC, Dinh J, Granados J, Flores-Villanueva PO. Joint effect of MCP-1 genotype GG and MMP-1 genotype 2G/2G increases the likelihood of developing pulmonary tuberculosis in BCG-vaccinated individuals. *PLoS One* 2010; 5:e8881.
28. Martinez FJ, Collard HR, Pardo A, Raghu G, Richeldi L, Selman M, *et al.* Idiopathic pulmonary fibrosis. *Nat Rev Dis Primers* 2017;3:17074.
29. Scabilloni JF, Wang L, Antonini JM, Roberts JR, Castranova V, Mercer RR. Matrix metalloproteinase induction in fibrosis and fibrotic nodule formation due to silica inhalation. *Am J Physiol Lung Cell Mol Physiol* 2005;288:L709–L717.
30. Al Shammari B, Shiomi T, Tezera L, Bielecka MK, Workman V, Sathyamoorthy T, *et al.* The extracellular matrix regulates granuloma necrosis in tuberculosis. *J Infect Dis* 2015;212:463–473.
31. Ordonez AA, Pokkali S, Sanchez-Bautista J, Klunk MH, Urbanowski ME, Kubler A, *et al.* Matrix metalloproteinase inhibition in a murine model of cavitary tuberculosis paradoxically worsens pathology. *J Infect Dis* 2019;219:633–636.
32. Urbanowski ME, Ihms EA, Bigelow K, Kubler A, Elkington PT, Bishai WR. Repetitive aerosol exposure promotes cavitary tuberculosis and enables screening for targeted inhibitors of extensive lung destruction. *J Infect Dis* 2018;218:53–63.
33. Rohlwink UK, Walker NF, Ordonez AA, Li YJ, Tucker EW, Elkington PT, *et al.* Matrix metalloproteinases in pulmonary and central nervous system tuberculosis—a review. *Int J Mol Sci* 2019;20:1350.
34. Xu Y, Wang L, Zimmerman MD, Chen KY, Huang L, Fu DJ, *et al.* Matrix metalloproteinase inhibitors enhance the efficacy of frontline drugs against *Mycobacterium tuberculosis*. *PLoS Pathog* 2018;14:e1006974.
35. Miow QH, Vallejo AF, Wang Y, Hong JM, Bai C, Teo FS, *et al.* Doxycycline host-directed therapy in human pulmonary tuberculosis. *J Clin Invest* 2021:131:e141895.
36. Winer A, Adams S, Mignatti P. Matrix metalloproteinase inhibitors in cancer therapy: turning past failures into future successes. *Mol Cancer Ther* 2018;17:1147–1155.
37. Vandenbroucke RE, Libert C. Is there new hope for therapeutic matrix metalloproteinase inhibition? *Nat Rev Drug Discov* 2014;13:904–927.
38. Muefong CN, Owolabi O, Donkor S, Charalambous S, Mendy J, Sey ICM, *et al.* Major neutrophil-derived soluble mediators associate with baseline lung pathology and post-treatment recovery in tuberculosis patients. *Front Immunol* 2021;12:740933.
39. Nwongbouwoh Muefong C, Owolabi O, Donkor S, Charalambous S, Bakuli A, Rachow A, *et al.* Neutrophils contribute to severity of tuberculosis pathology and recovery from lung damage pre- and posttreatment. *Clin Infect Dis* 2022;74:1757–1766.
40. Ndlovu LN, Peetluk L, Moodley S, Nhamoyebonde S, Ngoepe AT, Mazibuko M, *et al.* Increased neutrophil count and decreased neutrophil CD15 expression correlate with TB disease severity and treatment response irrespective of HIV co-infection. *Front Immunol* 2020; 11:1872.
41. Panteleev AV, Nikitina IY, Burmistrova IA, Kosmiadi GA, Radaeva TV, Amansahedov RB, *et al.* Severe tuberculosis in humans correlates best with neutrophil abundance and lymphocyte deficiency and does not correlate with antigen-specific CD4 T-cell response. *Front Immunol* 2017;8:963.
42. Wallis RS, Ginindza S, Beattie T, Arjun N, Likoti M, Edward VA, *et al.* Adjunctive host-directed therapies for pulmonary tuberculosis: a prospective, open-label, phase 2, randomised controlled trial. *Lancet Respir Med* 2021;9:897–908.
43. Alling DW, Bosworth EB. The after-history of pulmonary tuberculosis. VI. The first fifteen years following diagnosis. *Am Rev Respir Dis* 1960;81: 839–849.
44. Muefong CN, Sutherland JS. Neutrophils in tuberculosis-associated inflammation and lung pathology. *Front Immunol* 2020;11:962.
45. Stek C, Allwood B, Walker NF, Wilkinson RJ, Lynen L, Meintjes G. The immune mechanisms of lung parenchymal damage in tuberculosis and the role of host-directed therapy. *Front Microbiol* 2018;9:2603.

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

46. Moreira-Teixeira L, Stimpson PJ, Stavropoulos E, Hadebe S, Chakravarty P, Ioannou M, *et al.* Type I IFN exacerbates disease in tuberculosis-susceptible mice by inducing neutrophil-mediated lung inflammation and NETosis. *Nat Commun* 2020;11:5566.
47. Gopal R, Monin L, Torres D, Slight S, Mehra S, McKenna KC, *et al.* S100A8/A9 proteins mediate neutrophilic inflammation and lung pathology during tuberculosis. *Am J Respir Crit Care Med* 2013;188: 1137–1146.
48. Lovewell RR, Baer CE, Mishra BB, Smith CM, Sassetti CM. Granulocytes act as a niche for *Mycobacterium tuberculosis* growth. *Mucosal Immunol* 2021;14:229–241.
49. Daniels KJ, Irusen E, Pharaoh H, Hanekom S. Post-tuberculosis health-related quality of life, lung function and exercise capacity in a cured pulmonary tuberculosis population in the Breede Valley District, South Africa. *S Afr J Physiother* 2019;75:1319.
50. Pasipanodya JG, Miller TL, Vecino M, Munguia G, Bae S, Drewyer G, *et al.* Using the St. George Respiratory Questionnaire to ascertain health quality in persons with treated pulmonary tuberculosis. *Chest* 2007;132:1591–1598.
51. Visca D, Zampogna E, Sotgiu G, Centis R, Saderi L, D'Ambrosio L, *et al.* Pulmonary rehabilitation is effective in patients with tuberculosis pulmonary sequelae. *Eur Respir J* 2019;53:1802184.
52. Gupte AN, Selvaraju S, Gaikwad S, Mave V, Kumar P, Babu S, *et al.* Higher interleukin-6 levels and changes in transforming growth factor-β are associated with lung impairment in pulmonary tuberculosis. *ERJ Open Res* 2021;7:00390-2020.
53. Selman M, Pardo A. Revealing the pathogenic and aging-related mechanisms of the enigmatic idiopathic pulmonary fibrosis. an integral model. *Am J Respir Crit Care Med* 2014;189:1161–1172.
54. Adams TS, Schupp JC, Poli S, Ayaub EA, Neumark N, Ahangari F, *et al.* Single-cell RNA-seq reveals ectopic and aberrant lung-resident cell populations in idiopathic pulmonary fibrosis. *Sci Adv* 2020;6:eaba1983.
55. Tsukui T, Sun KH, Wetter JB, Wilson-Kanamori JR, Hazelwood LA, Henderson NC, *et al.* Collagen-producing lung cell atlas identifies multiple subsets with distinct localization and relevance to fibrosis. *Nat Commun* 2020;11:1920.
56. Lee HS, Hua HS, Wang CH, Yu MC, Chen BC, Lin CH. *Mycobacterium tuberculosis* induces connective tissue growth factor expression through the TLR2-JNK-AP-1 pathway in human lung fibroblasts. *FASEB J* 2019;33:12554–12564.
57. DiFazio RM, Mattila JT, Klein EC, Cirrincione LR, Howard M, Wong EA, *et al.* Active transforming growth factor-beta is associated with phenotypic changes in granulomas after drug treatment in pulmonary tuberculosis. *Fibrogenesis Tissue Repair* 2016;9:6.
58. Evans S, Butler JR, Mattila JT, Kirschner DE. Systems biology predicts that fibrosis in tuberculous granulomas may arise through macrophage-to-myofibroblast transformation. *PLoS Comput Biol* 2020; 16:e1008520.
59. Warsinske HC, DiFazio RM, Linderman JJ, Flynn JL, Kirschner DE. Identifying mechanisms driving formation of granuloma-associated fibrosis during *Mycobacterium tuberculosis* infection. *J Theor Biol* 2017;429:1–17.
60. Astuti T, Chozin IN, Damayanti NS, Nugrahenny D. The levels of pro-fibrotic cytokines in pulmonary tuberculosis with minimal and extensive lesions. *Lung India* 2018;35:204–208.
61. Dunlap MD, Prince OA, Rangel-Moreno J, Thomas KA, Scordo JM, Torrelles JB, *et al.* Formation of lung inducible bronchus associated lymphoid tissue is regulated by *Mycobacterium tuberculosis* expressed determinants. *Front Immunol* 2020;11:1325.
62. Hult C, Mattila JT, Gideon HP, Linderman JJ, Kirschner DE. Neutrophil dynamics affect *Mycobacterium tuberculosis* granuloma outcomes and dissemination. *Front Immunol* 2021;12:712457.
63. Canetti G. The tubercle bacillus in the pulmonary lesion of man: histobacteriology and its bearing on the therapy of pulmonary tuberculosis. New York, NY: Springer; 1955.
64. Nargan K, Naidoo T, Msimang M, Nadeem S, Wells G, Hunter RL, *et al.* Detection of *Mycobacterium tuberculosis* in human tissue via RNA in situ hybridization [preprint]. bioRxiv; 2023 [accessed 2024 Aug 16]. Available from: https://pubmed.ncbi.nlm.nih.gov/37873458/.
65. Hunter RL. Tuberculosis as a three-act play: a new paradigm for the pathogenesis of pulmonary tuberculosis. *Tuberculosis (Edinb)* 2016; 97:8–17.
66. Marakalala MJ, Raju RM, Sharma K, Zhang YJ, Eugenin EA, Prideaux B, *et al.* Inflammatory signaling in human tuberculosis granulomas is spatially organized. *Nat Med* 2016;22:531–538.
67. Seto S, Morimoto K, Yoshida T, Hiramatsu M, Hijikata M, Nagata T, *et al.* Proteomic profiling reveals the architecture of granulomatous lesions caused by tuberculosis and *Mycobacterium avium* complex lung disease. *Front Microbiol* 2019;10:3081.
68. Sawyer AJ, Patrick E, Edwards J, Wilmott JS, Fielder T, Yang Q, *et al.* Spatial mapping reveals granuloma diversity and histopathological superstructure in human tuberculosis. *J Exp Med* 2023;220:e20221392.
69. McCaffrey EF, Donato M, Keren L, Chen Z, Delmastro A, Fitzpatrick MB, *et al.* The immunoregulatory landscape of human tuberculosis granulomas. *Nat Immunol* 2022;23:318–329.
70. Gideon HP, Hughes TK, Tzouanas CN, Wadsworth MH 2nd, Tu AA, Gierahn TM, *et al.* Multimodal profiling of lung granulomas in macaques reveals cellular correlates of tuberculosis control. *Immunity* 2022;55: 827–846.e10.
71. Wells G, Glasgow JN, Nargan K, Lumamba K, Madansein R, Maharaj K, *et al.* A high-resolution 3D atlas of the spectrum of tuberculous and COVID-19 lung lesions. *EMBO Mol Med* 2022;14:e16283.
72. Wells G, Glasgow JN, Nargan K, Lumamba K, Madansein R, Maharaj K, *et al.* Micro-computed tomography analysis of the human tuberculous lung reveals remarkable heterogeneity in three-dimensional granuloma morphology. *Am J Respir Crit Care Med* 2021;204:583–595.
73. Stutz MD, Allison CC, Ojaimi S, Preston SP, Doerflinger M, Arandjelovic P, *et al.* Macrophage and neutrophil death programs differentially confer resistance to tuberculosis. *Immunity* 2021;54:1758–1771.e7.
74. Ramon-Luing LA, Olvera Y, Flores-Gonzalez J, Palacios Y, Carranza C, Aguilar-Duran Y, *et al.* Diverse cell death mechanisms are simultaneously activated in macrophages infected by virulent *Mycobacterium tuberculosis*. *Pathogens* 2022;11:492.
75. Nisa A, Kipper FC, Panigrahy D, Tiwari S, Kupz A, Subbian S. Different modalities of host cell death and their impact on *Mycobacterium tuberculosis* infection. *Am J Physiol Cell Physiol* 2022;323: C1444–C1474.
76. Amaral EP, Costa DL, Namasivayam S, Riteau N, Kamenyeva O, Mittereder L, *et al.* A major role for ferroptosis in *Mycobacterium tuberculosis*-induced cell death and tissue necrosis. *J Exp Med* 2019; 216:556–570.
77. Sousa J, Cá B, Maceiras AR, Simões-Costa L, Fonseca KL, Fernandes AI, *et al. Mycobacterium tuberculosis* associated with severe tuberculosis evades cytosolic surveillance systems and modulates IL-1β production. *Nat Commun* 2020;11:1949.
78. Stamm CE, Collins AC, Shiloh MU. Sensing of *Mycobacterium tuberculosis* and consequences to both host and bacillus. *Immunol Rev* 2015;264:204–219.
79. Stanley SA, Johndrow JE, Manzanillo P, Cox JS. The Type I IFN response to infection with *Mycobacterium tuberculosis* requires ESX-1-mediated secretion and contributes to pathogenesis. *J Immunol* 2007; 178:3143–3152.
80. Wassermann R, Gulen MF, Sala C, Perin SG, Lou Y, Rybniker J, *et al. Mycobacterium tuberculosis* differentially activates cGAS- and inflammasome-dependent intracellular immune responses through ESX-1. *Cell Host Microbe* 2015;17:799–810.
81. Koo IC, Wang C, Raghavan S, Morisaki JH, Cox JS, Brown EJ. ESX-1-dependent cytolysis in lysosome secretion and inflammasome activation during mycobacterial infection. *Cell Microbiol* 2008;10: 1866–1878.
82. Augenstreich J, Arbues A, Simeone R, Haanappel E, Wegener A, Sayes F, *et al.* ESX-1 and phthiocerol dimycocerosates of *Mycobacterium tuberculosis* act in concert to cause phagosomal rupture and host cell apoptosis. *Cell Microbiol* 2017;19:e12726.
83. Qiang L, Zhang Y, Lei Z, Lu Z, Tan S, Ge P, *et al.* A mycobacterial effector promotes ferroptosis-dependent pathogenicity and dissemination. *Nat Commun* 2023;14:1430.
84. Chai Q, Yu S, Zhong Y, Lu Z, Qiu C, Yu Y, *et al.* A bacterial phospholipid phosphatase inhibits host pyroptosis by hijacking ubiquitin. *Science* 2022;378:eabq0132.
85. Sapozhnikova NV, Skvortsova LA, Pavlova MV, Vishnevskiĭ BI, Otten TF, Vasil'eva SN, *et al.* Pulmonary tuberculosis caused by *Mycobacterium tuberculosis* of different genotypes [in Russian]. *Probl Tuberk Bolezn Legk* 2003:13–15.

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

86. Sun YJ, Lim TK, Ong AK, Ho BC, Seah GT, Paton NI. Tuberculosis associated with *Mycobacterium tuberculosis* Beijing and non-Beijing genotypes: a clinical and immunological comparison. *BMC Infect Dis* 2006;6:105.
87. Verma S, Bhatt K, Lovey A, Ribeiro-Rodrigues R, Durbin J, Jones-López EC, *et al.* Transmission phenotype of *Mycobacterium tuberculosis* strains is mechanistically linked to induction of distinct pulmonary pathology. *PLoS Pathog* 2019;15:e1007613.
88. Sharma A, Bloss E, Heilig CM, Click ES. Tuberculosis caused by *Mycobacterium africanum*, United States, 2004-2013. *Emerg Infect Dis* 2016;22:396–403.
89. de Jong BC, Adetifa I, Walther B, Hill PC, Antonio M, Ota M, *et al.* Differences between tuberculosis cases infected with *Mycobacterium africanum*, West African type 2, relative to Euro-American *Mycobacterium tuberculosis*: an update. *FEMS Immunol Med Microbiol* 2010;58:102–105.
90. Diarra B, Kone M, Togo ACG, Sarro YDS, Cisse AB, Somboro A, *et al.* *Mycobacterium africanum* (lineage 6) shows slower sputum smear conversion on tuberculosis treatment than *Mycobacterium tuberculosis* (lineage 4) in Bamako, Mali. *PLoS One* 2018;13:e0208603.
91. Namasivayam S, Diarra B, Diabate S, Sarro Y, Kone A, Kone B, *et al.* Patients infected with *Mycobacterium africanum* versus *Mycobacterium tuberculosis* possess distinct intestinal microbiota. *PLoS Negl Trop Dis* 2020;14:e0008230.
92. Silva ML, Cá B, Osório NS, Rodrigues PNS, Maceiras AR, Saraiva M. Tuberculosis caused by *Mycobacterium africanum*: knowns and unknowns. *PLoS Pathog* 2022;18:e1010490.
93. Ufimtseva EG, Eremeeva NI, Umpeleva TV, Vakhrusheva DV, Skornyakov SN. *Mycobacterium tuberculosis* load in host cells and the antibacterial activity of alveolar macrophages are linked and differentially regulated in various lung lesions of patients with pulmonary tuberculosis. *Int J Mol Sci* 2021;22:3452.
94. Kaplan G, Post FA, Moreira AL, Wainwright H, Kreiswirth BN, Tanverdi M, *et al. Mycobacterium tuberculosis* growth at the cavity surface: a microenvironment with failed immunity. *Infect Immun* 2003;71: 7099–7108.
95. Byrne AL, Marais BJ, Mitnick CD, Garden FL, Lecca L, Contreras C, *et al.* Chronic airflow obstruction after successful treatment of multidrug-resistant tuberculosis. *ERJ Open Res* 2017;3:00026-2017.
96. Kramnik I, Beamer G. Mouse models of human TB pathology: roles in the analysis of necrosis and the development of host-directed therapies. *Semin Immunopathol* 2016;38:221–237.
97. Leu J-S, Chen M-L, Chang S-Y, Yu S-L, Lin C-W, Wang H, *et al.* SP110b controls host immunity and susceptibility to tuberculosis. *Am J Respir Crit Care Med* 2017;195:369–382.
98. Ji DX, Witt KC, Kotov DI, Margolis SR, Louie A, Chevée V, *et al.* Role of the transcriptional regulator SP140 in resistance to bacterial infections via repression of type I interferons. *Elife* 2021;10:e67290.
99. Bhattacharya B, Xiao S, Chatterjee S, Urbanowski M, Ordonez A, Ihms EA, *et al.* The integrated stress response mediates necrosis in murine *Mycobacterium tuberculosis* granulomas. *J Clin Invest* 2021;131: e130319.
100. Krug S, Prasad P, Xiao S, Lun S, Ruiz-Bedoya CA, Klunk M, *et al.* Adjunctive integrated stress response inhibition accelerates tuberculosis clearance in mice. *mBio* 2023;14:e0349622.
101. Walter K, Kokesch-Himmelreich J, Treu A, Waldow F, Hillemann D, Jakobs N, *et al.* Interleukin-13-overexpressing mice represent an advanced preclinical model for detecting the distribution of antimycobacterial drugs within centrally necrotizing granulomas. *Antimicrob Agents Chemother* 2022;66:e0158821.
102. Heitmann L, Abad Dar M, Schreiber T, Erdmann H, Behrends J, McKenzie AN, *et al.* The IL-13/IL-4Ralpha axis is involved in tuberculosis-associated pathology. *J Pathol* 2014;234:338–350.
103. Subbian S, Tsenova L, O'Brien P, Yang G, Kushner NL, Parsons S, *et al.* Spontaneous latency in a rabbit model of pulmonary tuberculosis. *Am J Pathol* 2012;181:1711–1724.
104. Lin PL, Ford CB, Coleman MT, Myers AJ, Gawande R, Ioerger T, *et al.* Sterilization of granulomas is common in active and latent tuberculosis despite within-host variability in bacterial killing. *Nat Med* 2014;20: 75–79.
105. Lin PL, Rodgers M, Smith L, Bigbee M, Myers A, Bigbee C, *et al.* Quantitative comparison of active and latent tuberculosis in the cynomolgus macaque model. *Infect Immun* 2009;77:4631–4642.
106. Boucau J, Naidoo T, Liu Y, Dasgupta S, Jain N, Castillo JR, *et al.* A mouse model of TB-associated lung fibrosis reveals persistent inflammatory macrophage populations during treatment [preprint]. bioRxiv; 2024 [accessed 2024 Aug 16]. Available from: https://pubmed.ncbi.nlm.nih.gov/38895338/.
107. Ahookhosh K, Vanoirbeek J, Vande Velde G. Lung function measurements in preclinical research: what has been done and where is it headed? *Front Physiol* 2023;14:1130096.
108. Elkington P, Lerm M, Kapoor N, Mahon R, Pienaar E, Huh D, *et al.* In vitro granuloma models of tuberculosis: potential and challenges. *J Infect Dis* 2019;219:1858–1866.
109. Allwood BW, Byrne A, Meghji J, Rachow A, van der Zalm MM, Schoch OD. Post-tuberculosis lung disease: clinical review of an under-recognised global challenge. *Respiration* 2021;100:751–763.
110. Iakobachvili N, Leon-Icaza SA, Knoops K, Sachs N, Mazères S, Simeone R, *et al.* Mycobacteria–host interactions in human bronchiolar airway organoids. *Mol Microbiol* 2022;117:682–692.
111. Tezera LB, Bielecka MK, Chancellor A, Reichmann MT, Shammari BA, Brace P, *et al.* Dissection of the host-pathogen interaction in human tuberculosis using a bioengineered 3-dimensional model. *Elife* 2017;6: e21283.
112. Snider GL, Doctor L, Demas TA, Shaw AR. Obstructive airway disease in patients with treated pulmonary tuberculosis. *Am Rev Respir Dis* 1971;103:625–640.
113. Rees D, Murray J. Silica, silicosis and tuberculosis. *Int J Tuberc Lung Dis* 2007;11:474–484.
114. Sumpter C, Chandramohan D. Systematic review and meta-analysis of the associations between indoor air pollution and tuberculosis. *Trop Med Int Health* 2013;18:101–108.
115. Lai T-C, Chiang C-Y, Wu C-F, Yang S-L, Liu D-P, Chan C-C, *et al.* Ambient air pollution and risk of tuberculosis: a cohort study. *Occup Environ Med* 2016;73:56–61.
116. Fink DL, Oladele DA, Slack AJ, Odubela O, Musari-Martins T, Okechukwu A, *et al.* A multi-centre observational study of HIV, tuberculosis and risk of chronic lung disease in urban West Africa. *AIDS* 2022;36:1987–1995.
117. Mpagama SG, Msaji KS, Kaswaga O, Zurba LJ, Mbelele PM, Allwood BW, *et al.* The burden and determinants of post-TB lung disease. *Int J Tuberc Lung Dis* 2021;25:846–853.
118. Zifodya JS, Temu TM, Masyuko SJ, Nyale G, Kinuthia J, Page ST, *et al.* HIV, pulmonary infections, and risk of chronic lung disease among Kenyan adults. *Ann Am Thorac Soc* 2021;18:2090–2093.
119. Nightingale R, Chinoko B, Lesosky M, Rylance SJ, Mnesa B, Banda NPK, *et al.* Respiratory symptoms and lung function in patients treated for pulmonary tuberculosis in Malawi: a prospective cohort study. *Thorax* 2022;77:1131–1139.
120. Chamie G, Luetkemeyer A, Walusimbi-Nanteza M, Okwera A, Whalen CC, Mugerwa RD, *et al.* Significant variation in presentation of pulmonary tuberculosis across a high resolution of CD4 strata. *Int J Tuberc Lung Dis* 2010;14:1295–1302.
121. Alisjahbana B, Sulastri N, Livia R, Apriani L, Verrall AJ, Sahiratmadja E. Neutrophils and lymphocytes in relation to MMP-8 and MMP-9 levels in pulmonary tuberculosis and HIV co-infection. *J Clin Tuberc Other Mycobact Dis* 2022;27:100308.
122. Rambaran S, Naidoo K, Lewis L, Hassan-Moosa R, Govender D, Samsunder N, *et al.* Effect of inflammatory cytokines/chemokines on pulmonary tuberculosis culture conversion and disease severity in HIV-infected and -uninfected individuals from South Africa. *Front Immunol* 2021;12:641065.
123. Mave V, Gaikwad S, Barthwal M, Chandanwale A, Lokhande R, Kadam D, *et al.* Diabetes mellitus and tuberculosis treatment outcomes in Pune, India. *Open Forum Infect Dis* 2021;8:ofab097.
124. Arriaga MB, Araujo-Pereira M, Barreto-Duarte B, Nogueira B, Freire M, Queiroz ATL, *et al.* Regional Prospective Observational Research in Tuberculosis (RePORT)—Brazil Consortium. The effect of diabetes and prediabetes on antituberculosis treatment outcomes: a multicenter prospective cohort study. *J Infect Dis* 2022;225:617–626.

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026

125. Critchley JA, Restrepo BI, Ronacher K, Kapur A, Bremer AA, Schlesinger LS, *et al.* Defining a research agenda to address the converging epidemics of tuberculosis and diabetes: part 1: epidemiology and clinical management. *Chest* 2017;152:165–173.
126. Barreda NN, Arriaga MB, Aliaga JG, Lopez K, Sanabria OM, Carmo TA, *et al.* Severe pulmonary radiological manifestations are associated with a distinct biochemical profile in blood of tuberculosis patients with dysglycemia. *BMC Infect Dis* 2020;20:139.
127. Ronacher K, van Crevel R, Critchley JA, Bremer AA, Schlesinger LS, Kapur A, *et al.* Defining a research agenda to address the converging epidemics of tuberculosis and diabetes: part 2: underlying biologic mechanisms. *Chest* 2017;152:174–180.
128. Kumar NP, Fukutani KF, Shruthi BS, Alves T, Silveira-Mattos PS, Rocha MS, *et al.* Persistent inflammation during anti-tuberculosis treatment with diabetes comorbidity. *Elife* 2019;8:e46477.
129. Pavan Kumar N, Moideen K, Nancy A, Viswanathan V, Shruthi BS, Shanmugam S, *et al.* Plasma eicosanoid levels in tuberculosis and tuberculosis-diabetes co-morbidity are associated with lung pathology and bacterial burden. *Front Cell Infect Microbiol* 2019;9:335.
130. Kumar NP, Moideen K, Viswanathan V, Shruthi BS, Sivakumar S, Menon PA, *et al.* Elevated levels of matrix metalloproteinases reflect severity and extent of disease in tuberculosis-diabetes co-morbidity and are predominantly reversed following standard anti-tuberculosis or metformin treatment. *BMC Infect Dis* 2018;18:345.
131. Kumar NP, Moideen K, Nancy A, Viswanathan V, Shruthi BS, Sivakumar S, *et al.* Systemic RAGE ligands are upregulated in tuberculosis individuals with diabetes co-morbidity and modulated by anti-tuberculosis treatment and metformin therapy. *BMC Infect Dis* 2019;19:1039.
132. Shivakoti R, Dalli J, Kadam D, Gaikwad S, Barthwal M, Colas RA, *et al.* Lipid mediators of inflammation and resolution in individuals with tuberculosis and tuberculosis-diabetes. *Prostaglandins Other Lipid Mediat* 2020;147:106398.
133. Van Lettow M, Kumwenda JJ, Harries AD, Whalen CC, Taha TE, Kumwenda N, *et al.* Malnutrition and the severity of lung disease in adults with pulmonary tuberculosis in Malawi. *Int J Tuberc Lung Dis* 2004;8:211–217.
134. Podewils LJ, Holtz T, Riekstina V, Skripconoka V, Zarovska E, Kirvelaite G, *et al.* Impact of malnutrition on clinical presentation, clinical course, and mortality in MDR-TB patients. *Epidemiol Infect* 2011;139:113–120.
135. Hoyt KJ, Sarkar S, White L, Joseph NM, Salgame P, Lakshminarayanan S, *et al.* Effect of malnutrition on radiographic findings and mycobacterial burden in pulmonary tuberculosis. *PLoS One* 2019;14:e0214011.
136. Wang X, Luo L, Zhang D, Wang J, Ning X, Lin Y, *et al.* Factors associated with nutritional risk in patients with pulmonary tuberculosis and structural lung disease: a hospital-based cross-sectional study. *J Multidiscip Healthc* 2022;15:1799–1807.
137. Sinha P, Ponnuraja C, Gupte N, Prakash Babu S, Cox SR, Sarkar S, *et al.* Impact of undernutrition on tuberculosis treatment outcomes in India: a multicenter, prospective, cohort analysis. *Clin Infect Dis* 2023; 76:1483–1491.
138. Bhargava A, Bhargava M, Meher A, Benedetti A, Velayutham B, Sai Teja G, *et al.* Nutritional supplementation to prevent tuberculosis incidence in household contacts of patients with pulmonary tuberculosis in India (RATIONS): a field-based, open-label, cluster-randomised, controlled trial. *Lancet* 2023;402:627–640.
139. Bhargava A, Bhargava M, Meher A, Teja GS, Velayutham B, Watson B, *et al.* Nutritional support for adult patients with microbiologically confirmed pulmonary tuberculosis: outcomes in a programmatic cohort nested within the RATIONS trial in Jharkhand, India. *Lancet Glob Health* 2023;11:e1402–e1411.
140. Ozoh OB, Ojo OO, Dania MG, Dede SK, Adegboyega OA, Irurhe NK, *et al.* Impact of post-tuberculosis lung disease on health-related quality of life in patients from two tertiary hospitals in Lagos, Nigeria. *Afr J Thorac Crit Care Med* 2021;27: 10.7196/AJTCCM.2021.v27i2.135.
141. Vidyasagaran AL, Readshaw A, Boeckmann M, Jarde A, Siddiqui F, Marshall A-M, *et al.* Is tobacco use associated with risk of recurrence and mortality among people with TB? A systematic review and meta-analysis. *Chest* 2024;165:22–47.
142. Kim HJ, Baek S, Kim HJ, Lee JS, Oh YM, Lee SD, *et al.* The impact of smoking on airflow limitation in subjects with history of asthma and inactive tuberculosis. *PLoS One* 2015;10:e0125020.
143. Khosa C, Bhatt N, Massango I, Azam K, Saathoff E, Bakuli A, *et al.* Development of chronic lung impairment in Mozambican TB patients and associated risks. *BMC Pulm Med* 2020;20:127.
144. Guiedem E, Pefura-Yone EW, Ikomey GM, Nkenfou CN, Mesembe M, Yivala MM, *et al.* Cytokine profile in the sputum of subjects with post-tuberculosis airflow obstruction and in those with tobacco related chronic obstructive pulmonary disease. *BMC Immunol* 2020;21:52.
145. Fiske CT, Hamilton CD, Stout JE. Alcohol use and clinical manifestations of tuberculosis. *J Infect* 2009;58:395–401.
146. Kan CK, Ragan EJ, Sarkar S, Knudsen S, Forsyth M, Muthuraj M, *et al.* Alcohol use and tuberculosis clinical presentation at the time of diagnosis in Puducherry and Tamil Nadu, India. *PLoS One* 2020;15: e0240595.
147. Ragan EJ, Kleinman MB, Sweigart B, Gnatienko N, Parry CD, Horsburgh CR, *et al.* The impact of alcohol use on tuberculosis treatment outcomes: a systematic review and meta-analysis. *Int J Tuberc Lung Dis* 2020;24:73–82.
148. Martinez L, Gray DM, Botha M, Nel M, Chaya S, Jacobs C, *et al.* The long-term impact of early-life tuberculosis disease on child health: a prospective birth cohort study. *Am J Respir Crit Care Med* 2023;207: 1080–1088.
149. Montesi SB, Izquierdo-Garcia D, Désogère P, Abston E, Liang LL, Digumarthy S, *et al.* Type I collagen-targeted positron emission tomography imaging in idiopathic pulmonary fibrosis: first-in-human studies. *Am J Respir Crit Care Med* 2019;200:258–261.
150. Esmail H, Lai RP, Lesosky M, Wilkinson KA, Graham CM, Coussens AK, *et al.* Characterization of progressive HIV-associated tuberculosis using 2-deoxy-2-[^{18}F]fluoro-D-glucose positron emission and computed tomography. *Nat Med* 2016;22:1090–1093.
151. Malherbe ST, Shenai S, Ronacher K, Loxton AG, Dolganov G, Kriel M, *et al.*; Catalysis TBBC. Persisting positron emission tomography lesion activity and *Mycobacterium tuberculosis* mRNA after tuberculosis cure. *Nat Med* 2016;22:1094–1100.
152. Marshall H, Stewart NJ, Chan HF, Rao M, Norquay G, Wild JM. In vivo methods and applications of xenon-129 magnetic resonance. *Prog Nucl Magn Reson Spectrosc* 2021;122:42–62.
153. Migliori GB, Marx FM, Ambrosino N, Zampogna E, Schaaf HS, van der Zalm MM, *et al.* Clinical standards for the assessment, management and rehabilitation of post-TB lung disease. *Int J Tuberc Lung Dis* 2021;25:797–813.
154. Naidoo K, Perumal R. Advances in tuberculosis control during the past decade. *Lancet Respir Med* 2023;11:311–313.
155. Allwood BW, van der Zalm MM, Amaral AFS, Byrne A, Datta S, Egere U, *et al.* Post-tuberculosis lung health: perspectives from the First International Symposium. *Int J Tuberc Lung Dis* 2020;24:820–828.

Downloaded from https://academic.oup.com/ajrccm/article/210/8/979/8491180 by guest on 11 July 2026